

```
!pip install vaderSentiment pandas matplotlib seaborn
```

Collecting vaderSentiment

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Installing collected packages: vaderSentiment

Successfully installed vaderSentiment-3.3.2

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from vaderSentiment.vaderSentiment import SentimentIntensityAnalyzer
```

```
data = {
    "Review": [
        "The product is excellent and I love it.",
        "Amazing quality and very fast delivery.",
        "I am very happy with this product.",
        "The product is okay.",
        "It is an average product.",
        "The service was terrible.",
        "I am disappointed with the product.",
        "Very poor quality and bad experience.",
        "The product works as expected.",
        "Good product but delivery was slow."
    ]
}

df = pd.DataFrame(data)

df
```

Review

- | | |
|---|---|
| 0 | The product is excellent and I love it. |
| 1 | Amazing quality and very fast delivery. |
| 2 | I am very happy with this product. |
| 3 | The product is okay. |
| 4 | It is an average product. |
| 5 | The service was terrible. |
| 6 | I am disappointed with the product. |
| 7 | Very poor quality and bad experience. |
| 8 | The product works as expected. |
| 9 | Good product but delivery was slow. |

```
analyzer = SentimentIntensityAnalyzer()
```

```
def get_sentiment(text):  
    score = analyzer.polarity_scores(text)["compound"]  
  
    if score >= 0.05:  
        return "Positive"  
    elif score <= -0.05:  
        return "Negative"  
    else:  
        return "Neutral"
```

```
df["Sentiment"] = df["Review"].apply(get_sentiment)
```

```
df
```

Review Sentiment

- | | | |
|---|---|----------|
| 0 | The product is excellent and I love it. | Positive |
| 1 | Amazing quality and very fast delivery. | Positive |
| 2 | I am very happy with this product. | Positive |
| 3 | The product is okay. | Positive |
| 4 | It is an average product. | Neutral |
| 5 | The service was terrible. | Negative |
| 6 | I am disappointed with the product. | Negative |
| 7 | Very poor quality and bad experience. | Negative |
| 8 | The product works as expected. | Neutral |
| 9 | Good product but delivery was slow. | Positive |

```
sentiment_counts = df["Sentiment"].value_counts()

print(sentiment_counts)
```

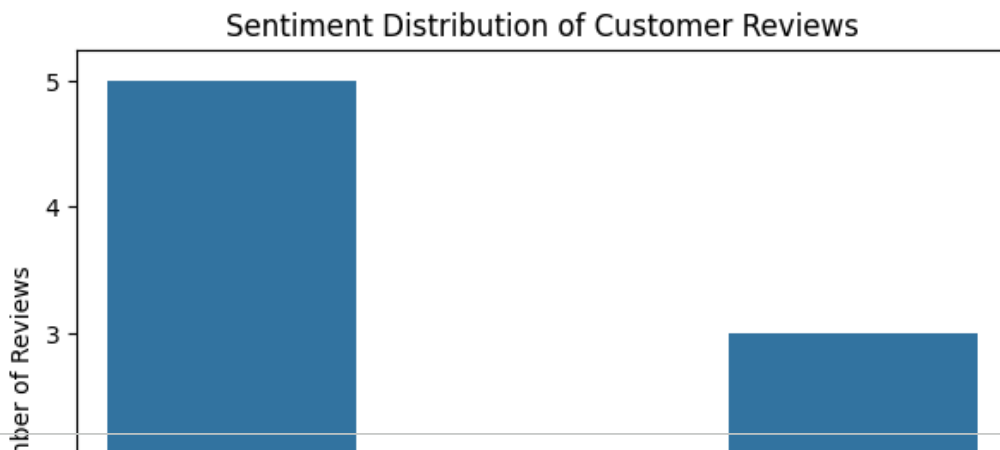
```
Sentiment
Positive    5
Negative    3
Neutral     2
Name: count, dtype: int64
```

```
plt.figure(figsize=(7,5))

sns.countplot(
    data=df,
    x="Sentiment"
)

plt.title("Sentiment Distribution of Customer Reviews")
plt.xlabel("Sentiment")
plt.ylabel("Number of Reviews")

plt.show()
```



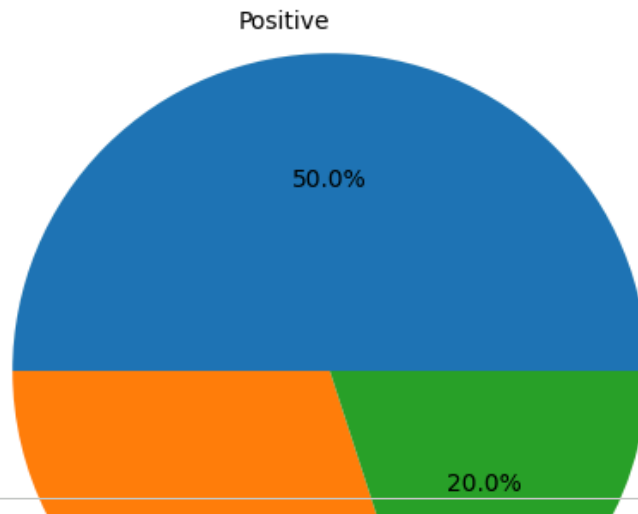
```
plt.figure(figsize=(6,6))

df["Sentiment"].value_counts().plot(
    kind="pie",
    autopct="%1.1f%%"
)

plt.title("Percentage of Sentiments")
plt.ylabel("")

plt.show()
```

Percentage of Sentiments



```
positive = (df["Sentiment"] == "Positive").sum()
negative = (df["Sentiment"] == "Negative").sum()
neutral = (df["Sentiment"] == "Neutral").sum()

print("Positive Reviews:", positive)
print("Negative Reviews:", negative)
print("Neutral Reviews:", neutral)

if positive > negative:
    print("\nOverall Opinion: Mostly Positive")
elif negative > positive:
    print("\nOverall Opinion: Mostly Negative")
else:
    print("\nOverall Opinion: Neutral/Mixed")
```

```
Positive Reviews: 5
Negative Reviews: 3
Neutral Reviews: 2
```

```
Overall Opinion: Mostly Positive
```

```
test_review = input("Enter a customer review: ")

sentiment = get_sentiment(test_review)

print("Review:", test_review)
print("Sentiment:", sentiment)
```

```
!pip install gradio
```

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```

✓ Conclusion

This project performs sentiment analysis on customer reviews using Natural Language Processing (NLP).

The VADER sentiment analyzer was used to classify reviews into three categories: Positive, Negative, and Neutral.

The results were visualized using bar charts and pie charts to understand the overall customer opinion.

This analysis can help organizations understand customer feedback, identify common opinions, and improve their products and services.

```

import gradio as gr

def analyze_review(review):
    if not review.strip():
        return "Please enter a review."

    sentiment = get_sentiment(review)

```

```
return f"Sentiment: {sentiment}"

demo = gr.Interface(
    fn=analyze_review,
    inputs=gr.Textbox(
        lines=5,
        placeholder="Enter a customer review here..."
    ),
    outputs=gr.Textbox(label="Result"),
    title="Sentiment Analysis of Customer Reviews",
    description="Enter a customer review to classify it as Positive, Negative, or Neutral",
    examples=[
        ["I absolutely love this product!"],
        ["The product is terrible and disappointing."],
        ["The product is okay."]
    ]
)

demo.launch(share=True)
```

Colab notebook detected. To show errors in colab notebook, set debug=True in launch()
* Running on public URL: <https://bb1bb9cc7915034122.gradio.live>

This share link is temporary and will last for up to 1 week (best effort). For free perm



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✓ Key Findings

- Customer reviews were successfully analyzed using NLP.
- Reviews were classified as Positive, Negative, or Neutral.

- The number of reviews in each category was calculated.
- Charts were created to visualize sentiment distribution.
- The overall customer opinion was identified from the results.

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  Downloading vaderSentiment-3.3.2-py2.py3-none-any.whl (125 kB)
```

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```

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Installing collected packages: vaderSentiment
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